

Computational Methods #3

Problem 1

Two ways of finding the volume of a parralelpiped are shown both using `std::vector` and `Eigen::Matrix3d`.

`std::vector<double>`

This was the less intuitive method. I created the `std::vector<double>` objects and the goal was to take a triple product $|(\vec{v}_a \times \vec{v}_b) \cdot \vec{v}_c|$. First I started with the cross product which I constructed element by element as I couldn't find a more reasonable method. then for the dot product I used the `std::inner_product`.

`Eigen::Matrix3d`

The goal here is to take the determinant of the matrix $(\vec{v}_a \ \vec{v}_b \ \vec{v}_c)$. This is done using the `.determinant()` method.

All of this is in the file `volume.cpp`.

Problem 2

First I convert the half lives to more reasonable decay times $\tau = \tau^{\text{half-life}} \ln(2)$. Now the rate equations are made simple by the fact that the ratios in the equation are 1 atom decaying into 1 atom for all process.

$$\dot{x}_b = -x_b/\tau_b, \quad \dot{x}_t = x_b/\tau_b - x_t/\tau_t, \quad \dot{x}_p = x_t/\tau_t$$

These Equations are written in a matrix form,

$$\frac{d}{dt} \begin{pmatrix} x_b \\ x_t \\ x_p \end{pmatrix} = \begin{pmatrix} -1/\tau_b & 0 & 0 \\ 1/\tau_b & -1/\tau_t & 0 \\ 0 & 1/\tau_t & 0 \end{pmatrix} \begin{pmatrix} x_b \\ x_t \\ x_p \end{pmatrix}$$

From here we identify the decay rate matrix Λ as,

$$\Lambda = \begin{pmatrix} -1/\tau_b & 0 & 0 \\ 1/\tau_b & -1/\tau_t & 0 \\ 0 & 1/\tau_t & 0 \end{pmatrix}$$

The solution to this equation of motion is

$$\vec{x}(t) = e^{\Lambda t} \vec{x}(0)$$

To convert the equation from number of particles to fraction just divide both sides by N_0 and then the vector $\vec{x}(t)$ becomes the fraction of each element as a function of time.

I created a function `getFractions.cpp` which solves the matrix equation and a main file `decay.cpp` which outputs the results to terminal.

Problem 3

In this problem the goal is to solve for the wavefunction

$$-\frac{\hbar^2}{2m} \psi''(x) + V(x) \psi(x) = E \psi(x)$$

Where the potential and the wavefunction are given by

$$V(x) \begin{cases} 0, & x < -2a \\ V_0, & -2a < x < -a \\ 2V_0, & -a < x < a \\ V_0, & a < x < 2a \\ 0, & x > 2a \end{cases}, \quad \psi(x) = \begin{cases} e^{ikx} + re^{-ikx} & x < -2a \\ ae^{in_1 kx} + be^{-in_1 kx} & -2a < x < -a \\ ce^{in_2 kx} + de^{-in_2 kx} & -a < x < a \\ fe^{in_1 kx} + ge^{-in_1 kx} & a < x < 2a \\ te^{ikx} & x > 2a \end{cases}$$

To solve for the coefficients we apply continuity and differentiability at all four boundaries.

Boundary 1:

$$\begin{aligned} e^{-2ika} + re^{2ika} &= ae^{-2in_1 ka} + be^{2in_1 ka} \\ e^{-2ika} - re^{2ika} &= n_1 ae^{-2in_1 ka} - n_1 be^{2in_1 ka} \end{aligned}$$

Boundary 2:

$$\begin{aligned} ae^{-in_1 ka} + be^{in_1 ka} &= ce^{-in_2 ka} + de^{in_2 ka} \\ n_1 ae^{-in_1 ka} - n_1 be^{in_1 ka} &= n_2 ce^{-in_2 ka} - n_2 de^{in_2 ka} \end{aligned}$$

Boundary 3:

$$\begin{aligned} ce^{in_2ka} + de^{-in_2ka} &= fe^{in_1ka} + ge^{-in_1ka} \\ n_2ce^{in_2ka} - n_2de^{-in_2ka} &= n_1fe^{in_1ka} - n_1ge^{-in_1ka} \end{aligned}$$

Boundary 4:

$$\begin{aligned} fe^{2in_1ka} + ge^{-2in_1ka} &= te^{2ika} \\ n_1fe^{2in_1ka} - n_1ge^{-2in_1ka} &= te^{2ika} \end{aligned}$$

In the previous equations I divided the derivative continuity equations by ik to remove the dangling momentum dimension and get all equations into the same dimension.

$$\begin{pmatrix} -e^{2ika} & e^{-2in_1ka} & e^{2in_1ka} & 0 & 0 & 0 & 0 & 0 & 0 \\ e^{2ika} & n_1e^{-2in_1ka} & -n_1e^{2in_1ka} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & e^{-in_1ka} & e^{in_1ka} & -e^{-in_2ka} & -e^{in_2ka} & 0 & 0 & 0 & 0 \\ 0 & n_1e^{-in_1ka} & -n_1e^{in_1ka} & -n_2e^{-in_2ka} & n_2e^{in_2ka} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & e^{in_2ka} & e^{-in_2ka} & -e^{in_1ka} & -e^{-in_1ka} & 0 & 0 \\ 0 & 0 & 0 & n_2e^{in_2ka} & -n_2e^{-in_2ka} & -n_1e^{in_1ka} & n_1e^{-in_1ka} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & e^{2in_1ka} & e^{-2in_1ka} & -e^{2ika} & -e^{2ika} \\ 0 & 0 & 0 & 0 & 0 & n_1e^{2in_1ka} & -n_1e^{-2in_1ka} & -e^{2ika} & -e^{2ika} \end{pmatrix} \begin{pmatrix} r \\ a \\ b \\ c \\ d \\ f \\ g \\ t \end{pmatrix} = \begin{pmatrix} e^{-2ika} \\ e^{-2ika} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

To solve this matrix equation, the scattering matrix is inverted and the coefficients are retrieved.

I created a file `getCoefficients.cpp` which handles the diagonalization process and a main file `scattering.cpp` which creates the output to the terminal.

Problem 4

The dynamics of the coupled harmonic oscillators is governed by the Lagrangian

$$L = \frac{1}{2} \dot{\vec{x}}^T M \dot{\vec{x}} - \frac{1}{2} \vec{x}^T K \vec{x}$$

Where each element of $\vec{x}$ is the displacement of the corresponding particle from equilibrium and the matrices M and K are given by

$$M = m \begin{pmatrix} 1 & & & & \\ & 3 & & & \\ & & 2 & & \\ & & & 1 & \\ & & & & 2 \end{pmatrix}, \quad K = k \begin{pmatrix} 1 & -1 & & & \\ -1 & 3 & -2 & & \\ & -2 & 4 & -2 & \\ & & -2 & 3 & -1 \\ & & & -1 & 1 \end{pmatrix}$$

Applying the Euler-Lagrange equations we get the equations of motion $M\ddot{\vec{x}} = -K\vec{x}$ which is a generalized eigenvalue problem. First use the ansatz $\vec{x}(t) = \vec{x}_0 e^{-i\omega t}$ and then convert it to a standard eigenvalue problem as follows,

$$\underbrace{(M^{-1/2} K M^{1/2})}_{\Omega^2} \underbrace{(M^{-1/2} \vec{x}_0)}_{\vec{y}_0} = \omega^2 \underbrace{(M^{-1/2} \vec{x}_0)}_{\vec{y}_0}$$

This equation is diagonalized to generate the eigenvectors $\vec{y}_i$. The solution to the dynamics is given by the equations in CH3. Since the initial velocities are zero then η_j are real and we can simplify the program using

$$\vec{y}(t) = \sum_j \eta_j \vec{y}_j \cos(\omega_j t), \quad \eta_j = \vec{y}_j \cdot \vec{y}(0)$$

The diagonalization is implemented and the final trajectories extracted. For the units to work out I diagonalize the matrices excluding an overall factor of k/m . This gives the eigenvalues units of $\omega_0^2 = k/m$. For the length units I have the inputPositions in cm and those units just propagate into the final answer as cm.

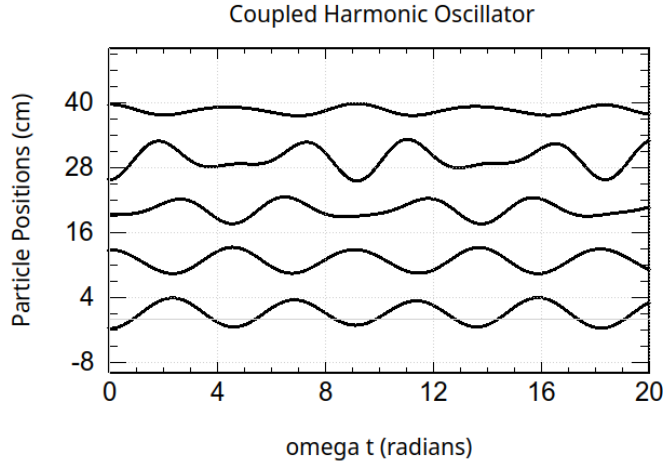


Figure 1: Position as a function of time for the five particles (particle 0 at the bottom and particle 4 at the top). I assumed the equilibrium positions of the particles to be 0 cm, 10 cm, 20 cm, 30 cm, 40 cm to make the plot easy to read.